

A Seminar Report
on
NEUROMORPHIC COMPUTING:
HARDWARE INSPIRED BY THE HUMAN BRAIN

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INTEGRATED COMPUTER SCIENCE AND ENGINEERING

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Certificate

This is to certify that the report entitled “**Neuromorphic computing : Hardware inspired by human brain**”, submitted by **Ammara Fairoz Sirkhot (24030332905059)**, is approved for partial fulfillment of the requirements for the award of the degree of **INTEGRATED COMPUTER SCIENCE AND ENGINEERING**.

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Abstract

Neuromorphic computing is a revolutionary approach to computer engineering that designs hardware and software to **mimic the structure and functionality of the human brain**. This paradigm moves beyond traditional **von Neumann architecture** by integrating processing and memory to enable highly parallel, **energy-efficient**, and real-time processing. Core foundations include **Spiking Neural Networks (SNNs)** and **Synaptic Plasticity**. This report details the history, hardware innovations (like **Memristors**), key players (Intel, IBM), and applications in Edge AI, Autonomous Systems, and Pattern Recognition. It concludes that neuromorphic computing is a strong contender for the future of efficient, adaptive, and real-time AI systems.

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Chapter 1

Introduction

Neuromorphic computing is an approach to engineering hardware and software that **mimics the structure and function of the human brain** to process information. Unlike traditional computers, which use a sequential, binary system, neuromorphic systems operate in a parallel, event-driven manner, offering significant advantages in **energy efficiency**, speed, and **adaptability**. These systems are implemented in analog, digital, or mixed-mode **VLSI** and prioritize robustness and learning by emulating the brain's distributed processing across small computing elements. This interdisciplinary field integrates biology, physics, mathematics, computer science, and electronic engineering to develop systems that emulate the brain's morphology and computational strategies.

1.1 Core Technical Foundations

Neuromorphic computing is an emerging paradigm that mimics the organizational principles and computational mechanisms of the human brain. It integrates concepts from neuroscience, computer engineering, and artificial intelligence to build highly efficient, brain-inspired hardware systems. The core technical foundations of neuromorphic computing include: Neuromorphic computing blends biological principles with advanced semiconductor engineering, enabling machines that learn, adapt, and process data like the human brain while consuming a fraction of the power of conventional computers. These foundations position neuromorphic systems as a crucial technology for the future of AI, robotics, and edge computing.

1.1.1 Spiking Neural Networks (SNNs)

Neuromorphic systems are based on SNNs, where neurons communicate using discrete electrical signals or "spikes," much like biological neurons. A neuron only activates and consumes power when an input event occurs, making the system incredibly energy-efficient. Information is often encoded in the timing and frequency of these spikes.

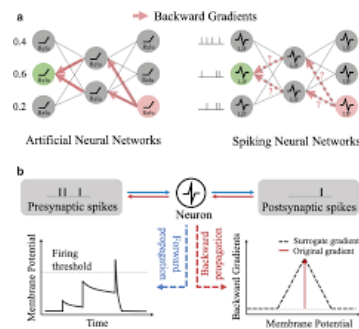


Figure 1.1: Spiking Neural Networks (SNNs)

1.1.2 Synaptic Plasticity

Emulating the brain's ability to learn and adapt, neuromorphic hardware implements **synaptic plasticity**, often through Spike Timing-Dependent Plasticity (STDP). This process adjusts the connection strength (weight) between neurons based on the relative timing of their spikes, enabling on-chip, continuous learning.

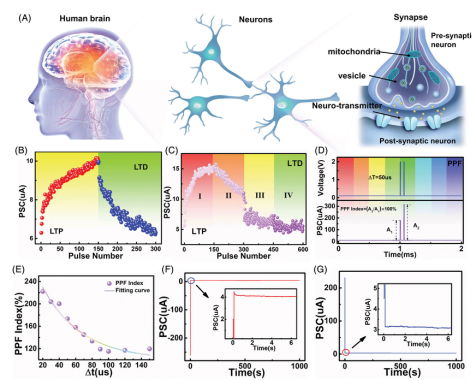


Figure 1.2: Synaptic Plasticity

Chapter 2

Brain-Inspired Architecture

2.1 Integrated Memory and Processing (Non-von Neumann Architecture)

To overcome the **von Neumann bottleneck**, memory components (synaptic weights) are **collocated with processing elements (neurons)**. This integrated approach allows for massive parallel computation, such as high-speed vector-matrix multiplication, and is crucial for the system's energy efficiency. Integrated memory and processing architectures (non-von Neumann) are a necessary evolution in computing, primarily driven by the need to overcome the **von Neumann bottleneck** and improve energy efficiency for data-intensive applications like AI and big data processing. By moving computation closer to or inside the memory, these designs significantly reduce data movement latency and energy consumption.

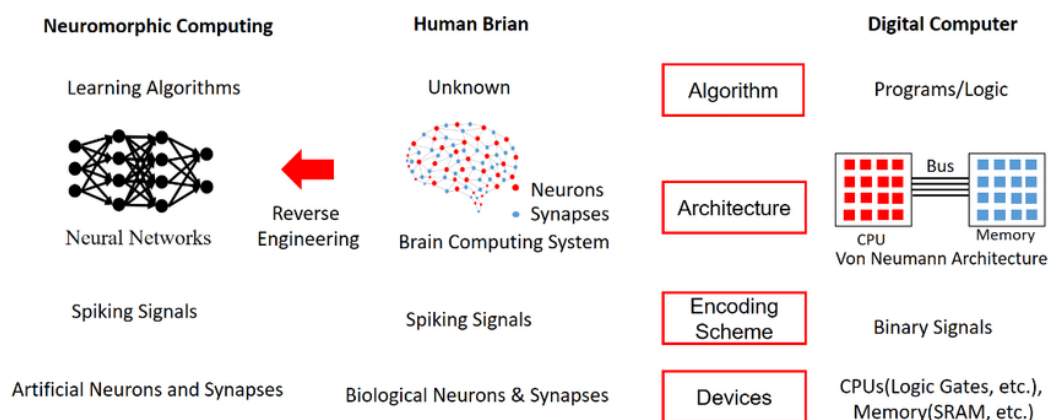


Figure 2.1: Comparison of Von Neumann and Neuromorphic Architectures

2.1.1 Neuromorphic Architecture vs. Von Neumann Architecture

The primary difference is that the **Von Neumann architecture separates the processing unit and memory**, creating a "bottleneck" when moving data back and forth. In contrast, **neuromorphic architecture integrates processing and memory** (in-memory computing), mimicking the human brain's parallel, event-driven operations to achieve superior energy efficiency and speed for AI tasks

Summary of Differences

Table 2.1: Architecture Comparison

Feature	Traditional (Von Neumann) Architecture	Neuromorphic Architecture (Brain-Inspired)
Structure	Separate CPU (Processor) and Memory (RAM/Storage)	Distributed Neuro-Synaptic Cores with integrated processing and memory.
Communication	Data moves back and forth between CPU and Memory via a Bus (The Von Neumann Bottleneck).	Event-Driven Spikes move directly between neighboring cores via a mesh network.
Processing	Clock-driven and sequential.	Asynchronous (no global clock) and massively parallel.
Efficiency	High power consumption.	Ultra-low power consumption (only active components draw power).

- **Efficiency:** Neuromorphic systems are significantly more energy-efficient (up to 1,000 to 10,000 times more efficient for certain tasks) because they are event-driven and avoid constant data shuttling.
- **Speed & Latency:** By avoiding the von Neumann bottleneck, neuromorphic chips can handle complex, real-time tasks like pattern recognition with lower latency than traditional processors.
- **Adaptability:** Neuromorphic hardware inherently supports learning and plasticity (the ability to change and adapt connections), which is a key feature of biological intelligence that is difficult to achieve efficiently in conventional architectures.

Chapter 3

Hardware Innovations and Materials

Hardware innovations in neuromorphic computing involve novel device architectures and new materials to mimic the brain's efficiency. Key advancements include memristive devices that act as artificial synapses, spintronic and photonic computing that use electron spin and light, and the use of two-dimensional (2D) materials like graphene for dense integration. These innovations move beyond traditional silicon CMOS, enabling more compact, energy-efficient, and scalable systems.

3.1 Memristors

A key component is the **memristor** (memory-resistor), a non-volatile electronic memory element whose resistance can be dynamically changed and remembered. These devices are ideal for mimicking biological synapses due to their small size, low power consumption, and ability to store multiple conductance states.

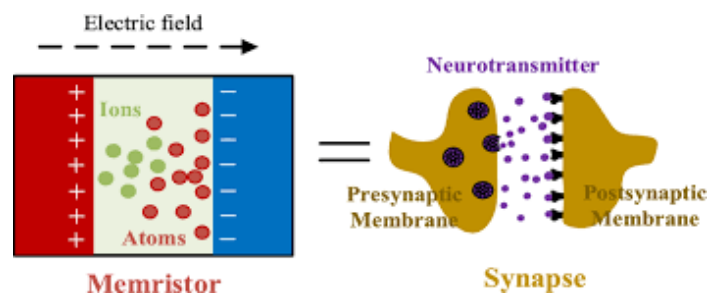


Figure 3.1: Memristor Function as an Artificial Synapse

3.1.1 Real-World Examples

The development of neuromorphic chips has been led by major technology companies and large-scale research projects.

Intel Loihi (Loihi 1 & 2)²



Figure 3.2: Intel Loihi (Loihi 1 & 2)²

Intel's Loihi chips are designed for on-chip learning and continuous adaptation, making them ideal for dynamic, real-world edge applications.³

- **Real-Time Optimization and Control:** Loihi has been demonstrated to solve complex **optimization problems** (like pathfinding for robots) hundreds of times more efficiently than conventional CPUs, crucial for **autonomous navigation** systems in robotics and drones.⁴
- **Sensor Fusion:** Loihi 2 has shown significant improvements in combining data from multiple sensors (e.g., radar, visual, and audio data) in real-time with ultra-low latency, which is essential for **autonomous vehicles** and complex **Industrial IoT** systems.⁵
- **Adaptive Prosthetics:** The chip's ability to perform on-chip learning allows **adaptive prosthetic limbs** to rapidly self-learn and adjust to a patient's unique gait and movements in real-time, improving usability and control.⁶

2. IBM TrueNorth⁷

Developed under the DARPA SyNAPSE program, TrueNorth was one of the first commercially viable, large-scale neuromorphic chips.⁸



Figure 3.3: IBM TrueNorth⁷

- **Cognitive Vision and Video Analysis:** Its massive array of digital neurons (1 million) is highly efficient for **real-time cognitive vision** tasks, such as simultaneous recognition and tracking of multiple objects in video streams at very low power (around 970 mW).¹⁰ *This is valuable for security and defense applications.*
- **Audio Localization and Classification:** TrueNorth has been used in prototypes for localizing and classifying sound sources, which is critical for **acoustic surveillance** and smart home systems.

3. BrainChip Akida

This neuromorphic System-on-Chip (NSoC) is focused on **Edge AI** for power-constrained devices.

- **Smart Vision for Edge Devices:** Akida excels at ultra-low-power tasks like **face recognition**, **gesture detection**, and **anomaly detection** in industrial settings, operating continuously ("always-on") in devices like smart cameras and wearables.¹¹



Figure 3.4: BrainChip Akida

- **Health Monitoring:** Prototypes include using Akida for efficient **seizure detection** in wearable or implantable medical devices, processing bio-signals locally with minimal power.¹²

Table 3.1: Leading Neuromorphic Chips

Chip Name	Developer	Key Features	Primary Focus
TrueNorth	IBM	1 Million Neurons, 256 Million Synapses.	Fixed pattern recognition.
Loihi 2	Intel	Up to 1 Million Neurons per chip, on-chip learning.	On-chip learning, robotics, complex SNNs.
SpiNNaker	U. of Manchester	500,000 ARM cores, large-scale brain simulation.	Neuroscience research, real-time biological modeling.

3.1.2 Emerging Concepts

Recent developments include proton-based artificial synapses, neuromorphic quantum computing, and organic artificial spiking neurons. Emerging concepts in neuromorphic hardware are focused on creating devices and architectures that implement **more complex, cognitive functions** than simple matrix multiplication, specifically by incorporating **time-dependent dynamics** and specialized memory structures.

Chapter 4

Software Frameworks and Programming

Software frameworks and programming models for neuromorphic computing are radically different from traditional CPU/GPU programming because they must account for the unique characteristics of the hardware: **event-driven communication**, **massive parallelism**, and **temporal dynamics** (time matters).

The field is broadly supported by three types of tools: **SNN Simulators**, **Hardware-Specific SDKs**, and **Generalized Frameworks**.

4.1 Lava Framework

Intel has spearheaded the open-source **Lava** software framework, which is platform-agnostic, allowing applications to run on both conventional CPUs/GPUs and specialized neuromorphic chips like Intel's Loihi. The **Lava Framework** is fundamentally about **software convergence and hardware abstraction** for neuromorphic computing. Its conclusion in the ecosystem is that it acts as the necessary **translation layer** that allows the development of complex, adaptive, and energy-efficient neuro-inspired applications, primarily for Intel's Loihi chips, while aiming for platform neutrality.

Modular Architecture

Lava allows developers to build neuromorphic applications using reusable modules called *processes*, making it flexible and scalable.

Hardware Agnostic

Although optimized for Intel Loihi, Lava programs can run on CPUs, GPUs, simulators, or future neuromorphic chips without changing the application logic.

Event-Driven Execution

Lava uses spike-based computation, where execution is triggered by events rather than continuous numeric processing, resulting in high energy efficiency.

Support for Learning Rules

It supports biologically inspired learning algorithms like:

- STDP (Spike-Timing-Dependent Plasticity)
- Reward-based learning
- Hebbian learning

4.2 Core Algorithmic Frameworks

Core algorithmic frameworks are basic models or structures used to design and solve problems using computer algorithms. These frameworks act like blueprints that guide how data is processed, stored, and used to make decisions. They help developers build reliable and efficient software systems.

Common Core Algorithmic Frameworks:

1. Divide and Conquer

Break a big problem into smaller parts, solve each part, and combine the results.

Example: Merge Sort, Quick Sort

2. Dynamic Programming

Solve complex problems by storing solutions to smaller problems to avoid repetition.

Example: Shortest path in graphs

3. Greedy Algorithms

Make the best choice at each step to get a quick, near-optimal solution.

Example: Finding minimum spanning tree

4. Graph Algorithms

Use nodes and connections to solve networking or relationship-based problems.

Example: Dijkstra's algorithm

5. Machine Learning Algorithms

Systems learn patterns from data and make predictions without being manually programmed.

Example: Neural networks

4.2.1 Supervised Classification

Training SNNs often involves methods analogous to supervised learning, such as error **backpropagation**, adapted for the discrete nature of spikes. Supervised classification in neuromorphic computing involves training spiking neural networks using labeled data, where each input is assigned a correct output category. The system learns patterns based on spike timing and then classifies new data by recognizing similar spike patterns. This approach enables fast, low-power decision-making similar to how the human brain identifies objects or sounds.

4.2.2 Unsupervised Classification

The core unsupervised learning mechanism is **Spike Timing-Dependent Plasticity (STDP)**, where synaptic weights are adjusted purely based on the relative firing times of pre- and post-synaptic neurons. Unsupervised classification in neuromorphic computing allows spiking neural networks to group and recognize patterns without using labeled data. The system learns on its own by detecting similarities in spike activity and forming clusters of related inputs. This makes the network capable of discovering hidden patterns or categories in data, just like the brain identifies new objects without prior knowledge.

4.3 Coding Templates

Implementation focuses on event-driven simulation, using models like the **Integrate-and-Fire** or **Leaky Integrate-and-Fire** models, the primary abstractions for biological neurons.

Types of Coding Templates:

1. **Rate Coding** Information is represented by the number of spikes fired in a given time. More spikes = stronger signal.
2. **Temporal Coding** The exact timing of spikes carries meaning. Early or late spikes represent different information.
3. **Population Coding** A group of neurons works together to represent data, improving accuracy.

4.3.1 Pseudo-Code for LIF Neuron

```
Function LIF_Neuron_Update(Voltage, Input_Current, Time_Step):
    Voltage = Voltage + (Input_Current - Leak_Current) * Time_Step
    If Voltage >= Threshold:
        Generate Spike
        Voltage = Reset_Potential
    Else:
        Voltage = Voltage
    Return Voltage
```

Chapter 5

Market Landscape and Applications

Neuromorphic computing is gaining attention due to its brain-like efficiency and low power usage. Many technology companies and research labs are investing in neuromorphic chips, sensors, and AI platforms. The market is growing because these systems can process data faster and learn in real time with very low energy.

5.1 Key Players and Products

The neuromorphic computing market is characterized by a few major players focusing on different segments, from research platforms to commercial edge IP. Several companies and research organizations lead the development of neuromorphic hardware and software. Their innovations are shaping the future of brain-inspired computing.

Table 5.1: Market Landscape

Company	Product	Focus
Intel	Loihi 2, Hala Point	Research chips for large-scale systems and optimization problems
IBM	TrueNorth, NorthPole	Early large-scale, low-power chips focusing on energy-efficient AI acceleration
BrainChip	Akida	Commercial IP for edge AI applications like industrial automation
SynSense	Speck, Xylo	Ultra-low-power chips for IoT and wearable devices

5.2 Applications and Use Cases

Neuromorphic computing is particularly well-suited for applications that require real-time processing and low power consumption:

- **Edge AI and IoT Devices:** Powering devices like smartphones, wearables, and sensors that process data locally, improving privacy and reducing latency.
- **Autonomous Systems:** Enhancing the real-time decision-making, navigation, and collision avoidance capabilities of robotics, drones, and self-driving cars.
- **Pattern Recognition:** Excelling in applications like natural language processing, speech recognition, and medical image analysis due to strong parallel processing capabilities.
- **Cybersecurity & Fraud Detection:** Detecting unusual patterns in data to identify cyberattacks or fraudulent financial activity rapidly.
- **Biomedical Engineering and Neural Prosthetics:** Assisting in brain–computer interfaces, artificial limbs, and devices that decode or stimulate neural signals for rehabilitation.
- **Smart Surveillance Systems:** Enabling real-time video analytics, threat detection, and abnormal behavior recognition without relying on cloud servers.
- **Aerospace and Defense:** Supporting autonomous spacecraft navigation, sensor fusion, adaptive targeting, and low-power mission-critical decision systems.
- **Smart Cities:** Managing huge sensor networks for traffic control, environmental monitoring, and energy-efficient infrastructure.
- **Industrial Automation:** Improving predictive maintenance, fault detection, and autonomous machinery operation in factories.
- **Brain Simulation and Cognitive Research:** Helping neuroscientists simulate and study human cognitive functions like learning, memory, and perception.

Chapter 6

Challenges and Future Outlook

6.1 Current Challenges

Despite its potential, neuromorphic computing faces challenges, including:

- **Software Maturity:** The lack of standardized software and programming models.
- **Learning Curve:** A steep learning curve for developers accustomed to traditional clock-driven architectures.
- **Performance Benchmarks:** A need for clear performance metrics to compare against conventional systems.
- **Ethical Considerations:** Advanced systems raise profound ethical questions regarding machine consciousness and legal ownership of generated work.

6.2 Future Scope

Future research focuses on expanding the capabilities and reach of neuromorphic systems:

- **Quantum Neuromorphic Computing:** Exploring quantum operations using neuromorphic systems.
- **Biointerfacing:** Developing organic artificial neurons for in-situ neuromorphic sensing.
- **Hybrid Architectures:** The future is likely hybrid, with specialized neuromorphic cores handling local, real-time sensing tasks, complementing large cloud-based AI models trained on conventional hardware.

Chapter 7

Conclusion

Neuromorphic computing offers a viable path toward energy-efficient, robust, and adaptable computational systems. By mimicking the brain, neuromorphic hardware successfully addresses the "**memory wall**" bottleneck inherent in von Neumann architectures, making it ideal for real-time sensory processing and AI applications. This transformative field is essential for the next generation of pervasive, sustainable, and adaptive computing. Despite challenges in manufacturing material variability and developing standardized SNN algorithms, the rapid growth and investment in this sector confirm its trajectory: Neuromorphic computing is the most promising path toward sustainable, adaptable, and genuinely brain-inspired Artificial Intelligence

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Appendix A

Summary of Neuromorphic Computing Hardware

A.1 Neuromorphic Concept Overview

- Neuromorphic computing mimics the structure and functionality of the human brain.
- Uses artificial neurons and synapses instead of traditional CPUs and GPUs.
- Focuses on event-driven parallel processing and low power consumption.

A.2 Key Hardware Components

1. **Artificial Neurons** – basic processing units that emulate biological neurons.
2. **Memristors** – devices that function like synapses and store memory.
3. **Spiking Neural Networks (SNNs)** – compute using spikes similar to brain signals.
4. **Crossbar Architectures** – enable parallel data paths for efficient computation.

A.3 Advantages of Neuromorphic Hardware

- Extremely low power consumption
- Highly parallel processing ability
- Faster learning and inference
- Closer biological realism compared to digital neural networks

A.4 Applications

- Robotics and autonomous devices
- Brain–machine interfaces
- Sensory perception systems (vision, hearing, touch)
- Edge and IoT devices requiring ultra-low power AI

Appendix B

Glossary

Neuron: A processing unit that performs computation using spikes.

Synapse: Connects neurons and stores memory strength (weight).

Memristor: A resistor with memory used to simulate biological synapses.

SNN: Spiking Neural Network – brain-inspired neural architecture.